

GenoFace — Progress & Next Steps (Plain English)

A simple guide to where the project stands, what we just finished, and what to do next.

1. What is this project?

We are trying to **predict eye colour and hair colour from DNA** using real people from the Personal Genome Project (PGP).

Each person gave:

- **Survey answers** about their eye and hair colour
- **Genotype files** (from 23andMe, AncestryDNA, etc.) uploaded to their PGP profile

We pull out specific SNPs (small DNA markers) that HlrisPlex-S — a well-known published tool — uses for prediction. Then we compare:

1. What HlrisPlex-S predicts from the DNA
2. What our own simple models predict
3. What the person actually reported in the survey

Skin colour is not part of this. The survey has no direct skin-colour question. We only have ethnicity as a rough guess, which is not good enough to train or evaluate a skin model properly.

2. General progress (the full journey so far)

Think of the project in four big phases:

Phase	What we did	Status
A. Get the data in shape	Downloaded two PGP surveys, cleaned messy answers, matched people across both surveys	Done
B. Find their DNA files	Visited PGP profiles online, found who has genotype files, downloaded and read them	Done
C. Build modeling datasets	Kept only people with enough SNPs ($\geq 80\%$ coverage) and a usable eye/hair label	Done
D. Match HlrisPlex-S labels	Renamed our labels so they use the exact same categories as HlrisPlex-S	Done
E. Predict & compare	Run HlrisPlex-S + train our own models, measure how often we get it right	Not started

How many people at each stage?

Raw surveys → ~1,000+ people with phenotype info
After joining → 1,046 people with both surveys
With genotype files → 86 successfully parsed
Eye modeling set → 71 people (final)
Hair modeling set → 62 people (final)
Both eye AND hair → 59 people overlap

Most people dropped off because they never uploaded DNA, their file could not be read, or they did not have enough of the SNPs we need.

3. Recent progress (what we just finished)

The most recent work was **label relabeling** so our ground truth matches HirisPlex-S exactly. Without this, comparing our results to HirisPlex-S would be unfair (like grading answers against the wrong answer key).

Eye colour — before and after

Our old label	HirisPlex-S label	What we did
blue	Blue	Renamed
brown	Brown	Renamed
hazel	Intermediate	Merged together
green	Intermediate	Merged together
gray	—	Removed (3 people)
amber	—	Removed (1 person)

Final eye dataset: 71 people (Blue: 18, Intermediate: 24, Brown: 29). These three groups are fairly balanced — good for training and testing.

Hair colour — before and after

Our old label	HirisPlex-S label	What we did
blonde	Blond	Renamed (HirisPlex spelling)
brown	Brown	Renamed
red	Red	Renamed
black	Black	Renamed
gray	—	Removed (8 people)
white	—	Removed (2 people)

Final hair dataset: 62 people (Brown: 48, Blond: 7, Black: 4, Red: 3). Brown dominates. Red is very rare (only 3 people). This matters a lot for how we evaluate models.

Files you have now

File	Purpose
eye_model_final.csv	71 people, 6 SNPs + eye label
hair_model_final.csv	62 people, 22 SNPs + hair label
csv-files/*_model_dataset.csv	Pre-relabel versions (kept as backup)
METHODS_LOG.md	Written record of the relabeling step

4. Data quirks to know before modeling

These affect what models make sense:

Eye (71 people, 6 SNPs)

- Only 2 people are missing one SNP (`rs1800407`). Easy to handle (drop those 2 rows, or fill in a default).
- Classes are balanced enough for normal cross-validation.

Hair (62 people, 22 SNPs)

- **Two SNPs are missing for almost everyone** (`rs1805009` and `rs312262906`). Consumer DNA chips often do not test these. You will train on ~20 usable SNPs, not 22.
- **Red hair: only 3 examples.** No model will learn red hair well from 3 people. Be honest about this in the report.
- **Brown: 48 out of 62 (77%).** A model that always guesses "Brown" would look ~77% accurate but would be useless for other colours.

59 people appear in both datasets — but eye and hair are separate prediction tasks; treat them independently unless you deliberately want a combined analysis.

5. What comes next (immediate steps)

Step 1 — Run HirisPlex-S on everyone (the benchmark)

- **What:** Feed each person's SNP genotypes into HirisPlex-S and record what it predicts.
- **Why:** HirisPlex-S is the published standard. It is the number you compare against. If your own model beats it, that is a result. If it matches HirisPlex-S, that shows you rebuilt something equivalent. If it is worse, you learn what the published tool adds.
- **How:** Use the [HirisPlex-S webtool](#) or a script that implements the same logic. Input the allele codes (e.g. `AG`, `GG`) from your CSV files.
- **Output:** A table per person: `participant_id | true_label | hirisplex_prediction | confidence/probability`

Step 2 — Train your own models

- **What:** Train simple classifiers on the same SNP data, using survey labels as the correct answer.

- **Why:** Shows you can build a prediction pipeline from scratch, not just use someone else's tool. Also lets you test whether simpler methods can get close to HlrPlex-S on your specific dataset.

Step 3 — Evaluate everything the same way

- **What:** Compare HlrPlex-S predictions and your model predictions against the true survey labels using the same tests.
- **Why:** Fair comparison only works if everyone is scored with the same rules on the same data splits.

6. Which models to train — and why

Given **71 eye samples** and **62 hair samples**, you do not have enough data for deep learning or complex models. Keep it simple and explainable.

Recommended models

Model	Eye (71, 6 SNPs)	Hair (62, ~20 SNPs)	Why use it
Logistic regression	Primary choice	Primary choice	Works well when you have few features and modest sample size. Easy to explain in an FYP ("each SNP gets a weight"). Less likely to overfit than fancier models.
Naive Bayes	Optional second model	Optional second model	Simple, fast, sometimes works well on genetic data where features are relatively independent. Good as a lightweight comparison.
Random forest	Optional comparison	Use with caution	Can capture non-linear patterns but with 62 samples and 20 features it can memorize the training data. Only include if you use proper cross-validation and compare honestly to logistic regression.
Neural networks	Do not use	Do not use	Not enough data. Will overfit and look good on training data but fail on test data. Hard to justify in a report.
Always-guess-majority baseline	Include	Must include for hair	Predicts the most common class every time (Brown for hair). Shows whether your real model is actually learning anything beyond "most people have brown hair."

What NOT to train

Task	Reason
Skin colour	No real skin-colour labels in the survey. Ethnicity is not a substitute. Skip entirely.
Red hair as a reliable class	3 examples is too few to trust any metric on Red alone. Report it separately with a clear warning.

7. How to evaluate — the WHAT and the WHY

The main rule: cross-validation

What: Split data into folds (e.g. 5 groups). Train on 4 groups, test on the 1 held-out group. Repeat 5 times so every person gets tested once. Average the results.

Why: With only 62–71 people, a single train/test split is unreliable — you might get lucky or unlucky depending on who lands in the test set. Cross-validation uses everyone for both training and testing (at different times), which is more stable on small datasets.

Use **stratified** folds — each fold keeps the same proportion of Blue/Brown/Intermediate (or Blond/Brown/Red/Black) as the full dataset.

Metrics to report

Metric	What it tells you	Why it matters for this project
Confusion matrix	A table showing predicted vs actual for every class	Easiest to understand visually. Shows exactly where mistakes happen (e.g. "we confuse Blond with Brown").
Overall accuracy	% of all predictions that were correct	Simple headline number. Misleading for hair because guessing Brown always gives ~77%.
Macro F1 score	Average performance across all classes, treating each class equally	More honest for hair. A model that ignores Red and Black still gets punished. Use this as your main hair metric.
Per-class precision & recall	For each colour: how often predictions were right, and how many real cases were caught	Essential for rare classes. For Red (n=3), report separately and say "too few to draw conclusions."
Compare to majority baseline	Is your model better than always guessing the most common class?	If logistic regression gets 80% on hair but majority baseline gets 77%, the model barely adds value.

Suggested evaluation flow

- For each trait (eye, hair):
1. Load eye_model_final.csv or hair_model_final.csv
 2. Handle missing SNPs (drop rows or columns with too many gaps)
 3. Encode genotypes as numbers (e.g. count of effect alleles per SNP)
 4. Run 5-fold stratified cross-validation:
 - a. HIrisPlex-S predictions (same folds, for fair comparison)
 - b. Logistic regression
 - c. Majority-class baseline
 - d. (Optional) Naive Bayes or Random Forest
 5. Collect confusion matrices and macro F1 for each method
 6. Make a summary table:

Method	Eye accuracy	Eye macro-F1	Hair accuracy	Hair macro-F1
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Majority baseline	...	...	...	...
Logistic regression	...	...	...	...
HIrisPlex-S	...	...	...	...

8. Realistic expectations

Be upfront in the FYP about what the numbers can and cannot show:

- **Eye colour (71 people, 3 balanced classes):** Reasonable to expect decent results. HlrPlex-S was designed for this. Macro F1 in the range of 0.6–0.8 would be plausible depending on label noise in self-reported surveys.
- **Hair colour (62 people, Brown-heavy):** Overall accuracy will look high mostly because of Brown. **Macro F1 is the number that matters.** It will likely be lower than accuracy. Red (n=3) and Black (n=4) results will be unstable — say so explicitly. Missing SNPs (`rs1805009` , `rs312262906`) may slightly reduce hair prediction quality vs a full HlrPlex-S run on complete data.
- **Self-reported labels are imperfect:** People describe hazel/green/blue differently. We merged some categories to match HlrPlex-S, but some mismatch between survey and DNA is expected — that caps how high any score can go.

9. Suggested order of work

Order	Task	Time estimate
1	Write a script to encode SNP genotypes from the CSV files	1–2 hours
2	Run HlrPlex-S on all 71 eye + 62 hair samples, save predictions	2–4 hours (manual webtool) or less with a script
3	Train logistic regression + majority baseline with 5-fold CV	2–3 hours
4	Build confusion matrices and summary comparison table	1–2 hours
5	Write up results in FYP — include limitations (small N, Red hair, missing SNPs, no skin)	Ongoing
6	(Optional) Add Naive Bayes or Random Forest as a second custom model	1 hour

10. One-paragraph summary

We built a complete data pipeline from raw PGP surveys and online genotype files down to two clean modeling datasets: **71 people for eye colour** (Blue / Intermediate / Brown) and **62 people for hair colour** (Blond / Brown / Red / Black), with labels aligned to HlrPlex-S. Skin colour is excluded because we lack proper labels. The next phase is prediction and evaluation: run HlrPlex-S as the benchmark, train simple models (mainly **logistic regression** with a **majority-class baseline** for comparison), and score everything with **stratified cross-validation**, **confusion matrices**, and **macro F1** — especially for hair, where class imbalance makes raw accuracy misleading. Red hair (only 3 people) should be reported with caution. The goal is a fair, honest comparison of what published HlrPlex-S achieves vs what we can build ourselves on real PGP data.